

My Iris Tracker

Project Brief V2.1

Iris growing, hybridising, seedling evaluation, and recordkeeping platform

Prepared for	Internal project planning and early product specification
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Version	V2.1, updated with mobile-first web app and PWA requirements
Status	Planning brief, not a final technical specification

Core idea

One sentence summary

A mobile-friendly iris platform that replaces fragmented spreadsheets and paper notes with a simple, structured system for tracking plants, parentage, crosses, seed development, photos, first flowering, seedling evaluation, future breeding ideas, and eventual sharing.

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1. Executive summary

This V2.1 brief reflects further feedback from an experienced iris grower and hybridiser, plus review of the British Iris Society registration form, the British Iris Society judges marking form, an existing hybridisation and seed development spreadsheet, and the American Iris Society Iris Wiki context.

The main conclusion is that My Iris Tracker should not be treated as a simple plant collection tracker. It should be designed as a specialist iris growing, hybridising, seedling evaluation, and recordkeeping platform. The first usable version can remain narrow and private, but the underlying structure should support the full breeding lifecycle from parent plants through to crosses, seed batches, seedlings, first flowering, evaluation, retention, naming, and possible registration support.

Primary product lesson

The app should connect records that are currently separate: plant records, parent photos, crosses, seed batches, germination results, seedlings, first flowers, evaluation notes, wants lists, and future cross ideas. This connected structure is where the app becomes more valuable than spreadsheets.

2. What has changed since V1

Learning	Implication for V2.1
From general tracker to specialist breeder platform	The app should still be easy to use, but it must support serious hybridising workflows rather than only garden inventory records.
Parentage becomes central	Pod parent and pollen parent should be structured records, not just free text. Unknown or bee-pollinated parentage must also be supported.
Seed development becomes a major module	The uploaded spreadsheet shows a clear need to track harvest, sowing, germination, repotting, planting out, first flower, retained seedlings, named seedlings, height, and notes.
Photos need categories	The app should support main flower photos, parent photos, seedling photos, whole plant photos, stem photos, foliage photos, disease photos, and registration-style photos.
Evaluation matters	The BIS judges marking form suggests future value in seedling evaluation fields covering foliage, disease resistance, growth, stems, buds, colour, form, substance, distinctiveness, and garden appeal.
Usability is non-negotiable	Many users may be older or uninterested in complex software. The app must hide complexity until needed and feel simple in everyday use.
Wants list and cross planner added	Users should be able to track varieties they want and plan future crosses before those crosses are attempted.
Help section added	Before wider testing, the app should include a plain-English

	Help section that acts like an instruction manual and glossary.
Future platform thinking refined	The app should be iris-specific first, but the underlying lifecycle model could later support other hybridised plants such as roses.

3. Reference material and key learning

The following materials should shape the V2.1 specification. They should not all be built into the first MVP, but they help define the direction of the product.

Reference	Learning
AIS Iris Wiki	The American Iris Society Iris Wiki is the public reference universe for irises, with cultivar information, images, classifications, parentage, genealogies, hybridizer context, and awards. The app should not try to replace it, but should allow users to link to relevant AIS Wiki records where useful.
AIS registration context	AIS is the official world registry for iris varieties. An iris can be registered and later introduced. This supports the idea of optional registration-support fields, not a mandatory registration workflow in the MVP.
BIS registration form	Provides useful descriptive fields: proposed name, seedling number, classification, stem height, bloom season, branch count, bud count, standards, style arms, falls, beard or signal, bloom size, fragrance, parentage, awards, and photographs.
BIS judges marking form	Shows that plant performance and garden evaluation matter, including foliage, disease resistance, growth, stem strength, branching, bud count, bloom sequence, colour, form, substance, distinctiveness, and garden appeal.
Existing hybridisation spreadsheet	Shows the practical lifecycle: season, pod parent, pollen parent, harvest date, pod number, sowing date, seed quantity, germination, repotting, final pots, planting into garden, first flower, retained seedlings, named seedlings, height, and free notes.
Dad feedback	Confirms the importance of parentage, parent photos, simplicity for older users, a wants list, future cross notes, and the possibility of adapting the system later for other hybridised plants.

4. Product vision and principles

The product should be designed around a serious but approachable vision:

Product vision

My Iris Tracker is a mobile-friendly iris recordkeeping platform for growers and hybridisers, designed to replace fragmented spreadsheets and paper notes with a simple, structured system for tracking plants, parentage, crosses, seed development, photos, first flowering, seedling evaluation, future breeding ideas, and later controlled sharing.

Core principles

- Start with one excellent private workflow for Dad, but structure the system as a multi-user platform from day one.
- Make everyday use simple, fast, and mobile-friendly.
- Keep advanced detail available, but do not force it into every screen.
- Treat parentage, crosses, seed batches, and seedlings as connected records rather than spreadsheet columns.
- Make photos part of the record structure, not an afterthought.
- Protect private breeding work by default.
- Support import and export so users never feel locked in.
- Use AIS and BIS terminology where relevant, while keeping language clear for non-technical users.
- Build in Help and guidance before broader testing.
- Keep the first public version iris-specific, even if the architecture could later support other plants.

5. Intended users

User type	Example	Main need
Primary early user	Dad and close private testers	Needs a better replacement for spreadsheets and paper notes. Must work on phone or tablet in the garden.
Serious iris hybridiser	Breeders and experienced growers	Needs cross records, parentage, seed development, seedling evaluation, photos, and long-term history.
Collector or enthusiast	Iris growers who do not breed heavily	Needs inventory, photos, locations, flowering records, wants list, and easy search.
Older or less technical user	A key usability group	Needs simple screens, clear labels, big buttons, minimal required fields, and plain-English Help.
Future society or club user	Possible later market	May need shared records, trial notes, public galleries, and group administration.
Future US user	Potential larger market	May need region settings, imperial and metric units, US-friendly terminology, and links to AIS resources.

6. Core workflow and information architecture

The system should follow the real iris breeding lifecycle rather than mirror one flat spreadsheet. This lifecycle should guide both the database design and the user interface.

Lifecycle model

Variety or plant record -> parentage -> cross -> seed pod -> seed batch -> sowing and germination -> seedling -> first flower -> evaluation -> retain, discard, name, or register-support record

This lifecycle can still be made simple in the app. A casual user may only create iris records and upload photos. A serious hybridiser can then open the advanced modules as needed.

7. Functional modules

7.1 Account, profile, and gardens

- Supabase Auth login from the first build.
- Each user has a profile and at least one garden.
- Every record belongs to a user and/or garden.
- Future support for club or shared gardens should be allowed by the database structure.

7.2 Iris records

The iris record is the central everyday object in the app. It should support simple and advanced views.

Area	Fields and requirements
Basic fields	Name, seedling code, type or classification, status, source, planted date, first flowered date, location, general notes.
Garden fields	Bed, row, pot, section, moved date, divided date, active or lost status, plant quantity if needed.
Descriptive fields	Colour notes, height, bloom season, fragrance, form, garden performance, and other characteristics.
Registration-support fields	Proposed name, alternative names, stem height, branch count, bud count, standards colour, style arms colour, falls colour, beard or signal colour, bloom size, awards, and photographs.
Visibility fields	Private by default, with future support for selected sharing, friends-only, and public records.

7.3 Garden locations

- Users can create garden locations such as beds, rows, pots, greenhouses, borders, trial areas, or holding areas.
- Locations should be optional but useful for searching and filtering.
- Plants may move over time, so movement history should be supported rather than simply overwriting location.

7.4 Notes and observations

- Each record should support dated notes.
- Common note types should include flowering, health, movement, pollination, seedling evaluation, photo note, and general note.
- The app should allow quick notes from the garden without forcing the user through a long form.
- Notes should be searchable.

7.5 Photos and media

Photos are a major part of the product. They should be private by default and linked to specific record types.

Photo type	Purpose
Main flower photo	Primary display image for an iris or seedling.
Parent photo	Photo of pod parent or pollen parent attached to a cross or parent record.
Seedling photo	Photo history for seedlings, including first flower and later performance.
Whole plant photo	Useful for vigor, height, plant habit, and garden appeal.
Stem photo	Useful for branch count, bud count, stem proportion, and evaluation.

Foliage or disease photo	Useful for trial notes, health issues, and disease resistance.
Registration-style photo	Optional photo set for name registration support or formal records.
Wanted variety photo	Photo attached to a wants list item or future cross idea.

7.6 Parentage and cross records

Parentage is a central feature. The app should allow parentage to be structured while still handling unknown or messy real-world cases.

- A cross should include season, pod parent, pollen parent, cross date if known, pod number, and notes.
- Pod parent and pollen parent can link to existing iris records where available.
- Unknown pollen parent should be supported.
- Bee pod or open-pollinated records should be supported.
- Parent photos should be supported, especially where a parent is not in the user's own garden records.
- Complex parentage should allow a free-text backup field.
- Future pedigree or genealogy view should be allowed by the data structure, but not required in the MVP.

7.7 Seed development and germination

The existing hybridisation spreadsheet strongly suggests a dedicated seed development module.

Stage	Fields and requirements
Harvest	Harvest date, quantity in pod, pod number, seed source notes.
Sowing	Sowing date, seed quantity, seed treatment, pot or batch reference.
Germination	First germination date, total germinated, germination percentage, notes.
Repotting	Repot date and plant quantity.
Final pots	Final pot date and plant quantity.
Planting out	Date planted into garden and plant quantity.
First flowering	First flower date and related photo or observation.
Final results	Quantity retained, quantity named, height, notes, and outcome.
Uncertainty	Labels possibly switched, dropped pots, unrecorded dates, unknown parentage, and other real-world notes must be allowed.

7.8 Seed treatment and experiment tracking

The spreadsheet shows A/B/C variants of crosses and notes about different seed treatments. The app should support treatment variants because this could become a valuable learning tool over time.

- Seed treatment method: dried then refrigerated, kept dry, chipped, fresh from pod, or custom.
- Treatment duration, such as six weeks or twelve weeks.
- Treatment notes.
- Ability to compare germination outcomes across treatment variants of the same cross.
- Keep this simple at first, with advanced comparison added later.

7.9 Seedling records and evaluation

Seedlings should become their own records once they need tracking beyond the seed batch. A seedling record should know which cross and seed batch it came from.

Area	Fields and requirements
Identity	Seedling code, cross reference, seed batch, generation, and status.
Photos	First flower photo, yearly photos, comparison photos, and parent photos.
Performance	First bloom date, repeat bloom notes, height, vigor, increase, disease notes.
Evaluation	Colour, form, substance, distinctiveness, stem strength, branching, bud count, bloom sequence, garden appeal, scent.
Outcome	Watch, retain, discard, name candidate, registered, introduced, or archived.
Notes	Free text for observations, mistakes, weather events, and breeder judgement.

7.10 Registration-support details

The app should not attempt to replace AIS or BIS registration systems. It can, however, help users gather and structure the information needed for a future registration application.

- Registration-support fields should be optional and hidden in an advanced area.
- Fields should broadly align with BIS/AIS-style descriptive information.
- Users should be able to export or copy the details when preparing an official application.
- Photographer and copyright notes should be supported for formal photo use.
- Registration support should be a later feature, not a requirement for the first MVP.

7.11 Wants list

The wants list is a useful feature for collectors and breeders. It should allow users to track varieties they want to obtain, research, or potentially use in future breeding.

- Wanted variety name.
- Photo or reference image.
- AIS Wiki or other reference link.
- Source, nursery, breeder, or contact.
- Reason wanted and desired traits.
- Possible use in future crosses.
- Priority and status, such as researching, wanted, acquired, no longer wanted.
- Ability to convert a wants list item into an iris record once acquired.

7.12 Future Cross Planner

The app should include a place to record possible future crosses before they are attempted.

- Potential pod parent and pollen parent.
- Desired outcome or trait goal.
- Parent photos and links to parent records.
- Flowering compatibility notes.
- Priority and timing notes.

- Free text reasoning.
- Convert planned cross into an actual cross when attempted.

7.13 Search, filters, import, and export

- Search should cover names, seedling codes, parents, locations, notes, and tags.
- Filters should include status, year, location, classification, parent, cross, seedling outcome, and visibility.
- CSV import should be planned early due to existing spreadsheet data.
- Import should allow column mapping and preview before saving.
- Export should be available for user trust and data portability.
- Future export options should include full garden export, breeding season export, cross export, and seedling list export.

8. Data model principles

The database should be designed for connected records and future multi-user access. Even if the first private version only has one user, every record must have ownership and privacy built in.

Table or area	Purpose
profiles	User profile, display name, region, preferences.
gardens	One or more gardens owned by a user or future organization.
locations	Beds, rows, pots, trial areas, greenhouse, holding areas.
iris_records	Named cultivars, seedlings, or individual plant records.
plantings	Location history, planted date, moved date, divided date, active or lost status.
photos	Private media linked to iris records, parents, crosses, seedlings, observations, or wants list items.
observations	Dated notes and measurements.
crosses	Pod parent, pollen parent, date, pod number, status, and notes.
seed_batches	Seeds harvested, treatment, sowing, germination, repotting, planting out.
seedlings	Individual seedling records linked to cross and seed batch.
evaluations	Scored or free-form performance notes for seedlings and trial plants.
wanted_items	Varieties wanted, photos, sources, and desired traits.
planned_crosses	Future cross ideas and breeding goals.
sharing_permissions	Future selected-user sharing and visibility controls.
help_articles	In-app Help content and guidance.
feedback_requests	Bug reports, support messages, feature suggestions.
drafts	Temporary unsynced or partially completed notes/photos planned for future offline-friendly workflows.

9. User experience requirements

The app must not feel like a complex database, even though the underlying structure will be detailed. This is especially important because some growers and hybridisers may be older or less comfortable with software.

- Mobile-first design, suitable for use outdoors in the garden.
- Tablet-friendly screens for older users who prefer a larger display.
- Large buttons and clear labels.
- Minimal required fields when adding a record.
- Advanced fields hidden until needed.

- Guided workflows for crosses, seed batches, and seedlings.
- Quick actions for Add photo, Add note, Record flowering, Record pollination, and Mark as retained or discarded.
- Plain language instead of technical database language.
- Strong search and sensible filters.
- Clear status labels and visual cues.
- Import and export should be reassuring and easy to understand.
- Error messages should explain what to do next.

UX rule

If a user wants to add one photo and one quick note while standing in the garden, they should be able to do it in a few taps without seeing the advanced database structure.

10. Mobile-first web app and PWA requirements

The first proper version of My Iris Tracker should be built as a mobile-first web app. The browser version should already feel good on a phone before any native Apple or Android app is considered. Native apps may be explored later, but the core product should first work extremely well in Safari and Chrome on mobile, tablet, and desktop.

Mobile-first product rule

The app should feel like a simple garden tool on a phone, not a spreadsheet squeezed into a small screen. If a user is standing outside and wants to add a photo, add a note, or find a record quickly, the interface should make that feel obvious and low-effort.

10.1 Phone and tablet experience

- Design for phone use first, especially while the user is outdoors in the garden.
- Use large buttons, clear labels, generous spacing, and touch-friendly controls.
- Keep navigation clear and consistent, with key actions such as Add iris, Add note, Add photo, Search, and Help easy to find.
- Avoid tiny spreadsheet-style tables on mobile. Use cards, grouped sections, and simple lists instead.
- Support a good tablet layout for users who prefer a larger screen.
- Ensure the app works well in Safari and Chrome, as these are the likely browsers on iPhone, iPad, Android, and desktop.
- Make the simplest daily actions possible in a few taps, especially adding a photo, adding a note, and searching for an iris.

10.2 Forms and workflows

- Build clean, touch-friendly forms with minimal required fields.
- Split advanced workflows into guided steps rather than long intimidating forms.
- Use sensible defaults and optional advanced sections.
- Make the Add note flow simple and fast, with common note types available as quick options.
- Make photo upload fast and obvious, using standard browser camera and photo upload features.
- Allow a user to take or upload a photo from the phone without needing a complex media workflow.

- Where possible, let users save simple drafts if signal is poor, but do not overbuild full offline mode in the early version.

10.3 Technical front-end principles

- Build fully responsive mobile-first screens from the start.
- Avoid relying on desktop-only layouts, hover-only controls, wide tables, or dense grid views.
- Build reusable UI components for records, notes, photos, status labels, form sections, and quick actions.
- Keep business logic separate from UI components where possible, so the app can later be wrapped or adapted more easily.
- Store data in Supabase rather than relying on local-only storage.
- Use local storage or browser storage only for temporary drafts, preferences, or offline-friendly improvements where appropriate.
- Avoid web-only hacks that may fail inside a future iOS or Android app wrapper.
- Plan the codebase so a future Capacitor-style native wrapper remains possible without rebuilding the whole product.

10.4 PWA readiness

The early product should be planned as an installable Progressive Web App, even if App Store and Google Play versions are not part of the MVP. This gives users an app-like home-screen experience while keeping development simpler.

- Add a proper web app manifest.
- Create suitable icons for phone, tablet, and browser use.
- Set appropriate app name, short name, theme colour, and display settings.
- Make the app installable to a phone or tablet home screen where supported.
- Use responsive layouts that work cleanly when launched from the home screen.
- Plan offline drafts carefully, but do not attempt full offline sync until the core online workflows are stable.
- Keep all important records synced to Supabase once saved, so users are not dependent on one device.

10.5 Future native app path

The first build should not be a native Apple or Android app. However, the web app should be built cleanly enough that a future iOS and Android version could be created using a web-to-native wrapper if the product proves itself. This means the browser app should avoid shortcuts that would block future packaging, especially around navigation, file upload, authentication, and local storage.

11. Help and user guidance

A Help section should be introduced around the time the app expands beyond private MVP testing into additional user testing. It is not required for the first private build, but it should be planned as a core support feature.

- In-app Help section acting as a practical instruction manual.
- Getting started guide for first-time users.
- Step-by-step articles for adding an iris, adding a photo, creating a cross, tracking seed batches, creating seedlings, using the wants list, and planning future crosses.

- Glossary of iris and hybridising terms, such as pod parent, pollen parent, bee pod, seedling, standards, falls, beard, signal, style arms, classification, registered, and introduced.
- Contextual help icons on advanced fields.
- Plain-English wording aimed at non-technical users.
- Still stuck contact option beneath help articles.
- Feature request link from the Help area.
- Potential feedback prompt: Was this helpful?

12. Security, privacy, and data protection

Security and data protection must be designed into the app from the beginning, not added later. This matters because breeding records, photos, personal notes, and future public sharing settings all need clear ownership and privacy controls.

Area	Requirement
Authentication	Use Supabase Auth with email/password and verified user accounts. Add stronger account controls later as needed.
Row Level Security	Enable RLS on every user-owned table. Users must only access their own data unless an explicit sharing rule allows access.
Ownership	Every garden, record, photo, observation, cross, seed batch, seedling, wanted item, and planned cross must have clear ownership.
Private by default	All records and photos should be private unless the user actively shares or publishes them.
Photo storage	Use private Supabase Storage buckets, not public image buckets, for user-uploaded photos.
Secrets	Never expose service role keys or sensitive values in frontend code. Use server-only code for sensitive operations.
Data minimisation	Only collect what the app needs: email, profile name, garden records, photos, and user preferences. Avoid precise addresses unless genuinely required later.
Export and deletion	Plan for data export, photo deletion, record deletion, and future account deletion.
Auditability	Later versions should log important actions such as sharing, deleting, importing, exporting, or changing visibility.
Legal readiness	Before wider public launch, prepare privacy policy, terms, cookie policy, refund policy, public content terms, and data deletion policy.

13. Environments, testing, and deployment

The app should be built with a safe workflow so changes are tested away from live data.

Environment	Purpose
Local development	Claude Code and the developer work locally on new features.
Preview deployments	Vercel preview links allow individual branches or changes to be tested before release.
Staging	A persistent test version with test data for larger changes and Dad or beta user review.
Production	The real live app with real records and photos.
Database separation	Use separate local, staging, and production databases before

	inviting wider testers.
Backups	During private MVP, CSV exports may be enough. Once real long-term data is being stored, use automated database backups and a photo backup strategy.
Free tiers	Vercel and Supabase free tiers should be enough for private MVP and very small testing, but a commercial version will need paid plans and stronger operations.

14. Build phases

Phase	Scope
Phase 0: Discovery	Collate current spreadsheets, confirm fields, gather examples, define terminology, map current workflows, and prepare import strategy.
Phase 1: Private MVP	Login, profile, garden, locations, iris records, notes, photos, search, basic filters, CSV export, mobile-first screens, touch-friendly forms, and simple garden use.
Phase 2: Breeding MVP	Cross records, parentage, pod parent, pollen parent, unknown parent, seed batches, seed development, germination, seedlings, and basic outcomes.
Phase 3: Evaluation, PWA, and garden workflow	Seedling evaluation, first flowering, keep/watch/discard status, quick garden actions, better filters, improved photo workflows, app manifest, icons, and PWA readiness.
Phase 4: Help and wider testing readiness	Help section, glossary, contextual prompts, feedback/contact forms, staging environment, onboarding, and selected beta users.
Phase 5: Sharing and community testing	Private/shared/public visibility, selected-user sharing, friend or follow connections, public pages, and shareable links.
Phase 6: Commercial readiness	Subscriptions, legal documents, billing, refund policy, user support processes, account deletion, export tools, and public launch preparation.
Phase 7: Future expansion	Potential club accounts, trial garden tools, deeper AIS Wiki linking, advanced pedigree views, and possible future versions for other hybridised plants.

15. Commercial readiness

Commercial features are not part of the early MVP, but they should be planned so the app does not need to be rebuilt later.

- Future paid plans could separate casual growers from serious breeders.
- Possible launch structure: free plan, hobby plan, breeder plan, and later club or society plan.
- Pricing should be tested with real users. An early hypothesis is around \$49 per year for normal growers and \$79 to \$99 per year for serious breeders if the tool genuinely replaces spreadsheets and adds valuable breeding features.
- Before charging users, the app needs privacy policy, terms, cookie policy, refund policy, billing terms, public content terms, and clear cancellation/data export handling.
- A contact section should include general questions, bug reports, feature suggestions, import help, billing questions, privacy requests, and club enquiries.

16. Future adaptability beyond irises

The first version should stay focused on irises. That focus is important because iris-specific language, parentage, classification, bloom descriptions, and registration-support fields create the value. However, the underlying lifecycle model could later be adapted to other commonly hybridised plants, such as roses.

Future architecture principle
Build the core as plant -> parentage -> cross -> seed batch -> germination -> seedling -> evaluation -> selection. Then keep iris-specific fields modular, so other plant-specific versions could be explored later without weakening the iris product now.

17. Information still needed from Dad

Before finalising the technical build specification, the following information should be gathered from Dad.

Information needed	Why it matters
Current spreadsheets	Copies of all active spreadsheets, including messy versions. The messy reality is useful because it shows the real workflow.
Master field list	A full list of fields currently tracked, plus fields he wishes he tracked.
Essential vs optional	Mark each field as essential, useful, nice to have, or not needed.
Example records	One named iris, one seedling, one successful cross, one failed cross, one seed batch, one retained seedling, one discarded seedling, one moved plant, and one lost plant.
Naming and coding rules	How crosses, pod numbers, seed batches, seedlings, and named varieties are currently coded.
Photo folders	Examples of current iris photos and how they are organised.
Parentage habits	How often parents are known, unknown, bee-pollinated, or complex.
Seed treatment habits	Whether treatments like refrigeration, dry storage, chipping, or other methods should be standard options.
Garden use	Phone or tablet preference, signal reliability, and whether quick notes or voice notes would help.
Other growers	What Dad thinks other iris growers, especially US growers, would need that he does not currently track.
Help needs	Which parts of the workflow he thinks older or less technical users would find confusing.

18. Claude Code build guidance

The following instruction should be used as the foundation for Claude Code when moving from planning into implementation.

Claude Code instruction
Build My Iris Tracker as a secure, multi-user, mobile-friendly iris growing and hybridising platform. The first version will be used privately, but the database must support future accounts, sharing, and commercial use. Every user-owned table must include clear ownership and Row Level Security. Records and photos must be private by default. Build the MVP around simple iris records, garden locations, notes, photos, search, export, mobile-first screens, touch-friendly forms, fast browser photo upload, and reusable components, but design the data model so later phases can add parentage, crosses, seed batches, germination, seedlings, evaluation, wants lists, future cross planning, Help, sharing, and subscriptions without major restructuring.

- Do not build the whole platform in one pass.
- Start with schema planning and confirm the model before coding screens.
- Build in small stages with testable outputs.
- Use clear migration files for database changes.
- Keep UI simple, mobile-first, touch-friendly, and usable in Safari and Chrome.
- Avoid exposing database complexity to users.
- Use private storage and secure server-only operations for sensitive actions.
- Include import/export planning early.
- Add a proper app manifest and icon set for PWA readiness. Do not rely on desktop-only layouts or web-only hacks that would make a future app wrapper difficult.
- Use staging and preview environments before any wider testing.

19. References consulted

The following sources and materials informed this V2.1 brief. Links are included where the material is public.

Source	Use in brief	Reference
American Iris Society Iris Wiki	Public iris encyclopedia and classification reference	https://wiki.irises.org/
AIS Iris Wiki: About the Iris Encyclopedia	Context on the encyclopedia as a major iris database and reference	https://wiki.irises.org/Main/InfoAboutTheEncyclopedia
American Iris Society registration information	Context on registration, introduction, identifying characteristics, image upload, and registrar review	https://www.irises.org/hybridizers-growers/iris-registration/
AIS Registration and Checklists	Context on AIS as official world registry and distinction between registered and introduced	https://www.irises.org/about-ais/registration-checklists/
British Iris Society: Registering a new iris	Guidance on registration fields, naming, parentage, and photographs	https://www.britisshirissociety.org.uk/registering-a-new-iris/
BIS Application to Register an Iris Name	User-supplied PDF registration form with descriptive fields and photo permission wording	Uploaded document: RegistrationForm.pdf
BIS Judges Marking Form	User-supplied spreadsheet showing assessment criteria and scoring fields	Uploaded document: Judges Marking Form.xlsx
Dad hybridisation spreadsheet	User-supplied spreadsheet showing real cross, seed, germination, repotting, planting, first flower, and final result workflow	Uploaded document: GI Iris Hybridisation - Seed development database.xlsx

Appendix A: MVP versus later scope

Feature	Timing	Reason
Login and private profile	MVP	Required to establish secure ownership from the start.
Iris records	MVP	Core replacement for spreadsheet records.
Garden locations	MVP	Needed for practical garden use.
Notes and observations	MVP	Needed to replace paper notes.
Photo upload	MVP	Core value, with fast standard browser camera/photo upload, storage control, and compression.
CSV export	MVP	Builds trust and protects user data.
Mobile-first responsive UI	MVP	The first proper version must feel good on phone and tablet, not just desktop.
PWA manifest and icons	Phase 3	Allows app-like home-screen use before native apps are considered.

CSV import	Early after MVP	Likely important due to existing spreadsheets, but can be staged.
Cross records	Phase 2	Essential breeder feature.
Seed batches and germination	Phase 2	Based directly on current spreadsheet workflow.
Seedling evaluation	Phase 3	Needed for serious hybridising and judging-style assessment.
Wants list	Phase 3	Useful for collectors and future breeders.
Future cross planner	Phase 3	Useful for planning, not just recording.
Help section	Phase 4	Needed before wider testers use detailed features.
Sharing and public pages	Phase 5	Useful later, but avoid building social features too early.
Subscriptions	Phase 6	Only after strong user validation.
Other plant versions	Future	Possible only after iris version is proven.